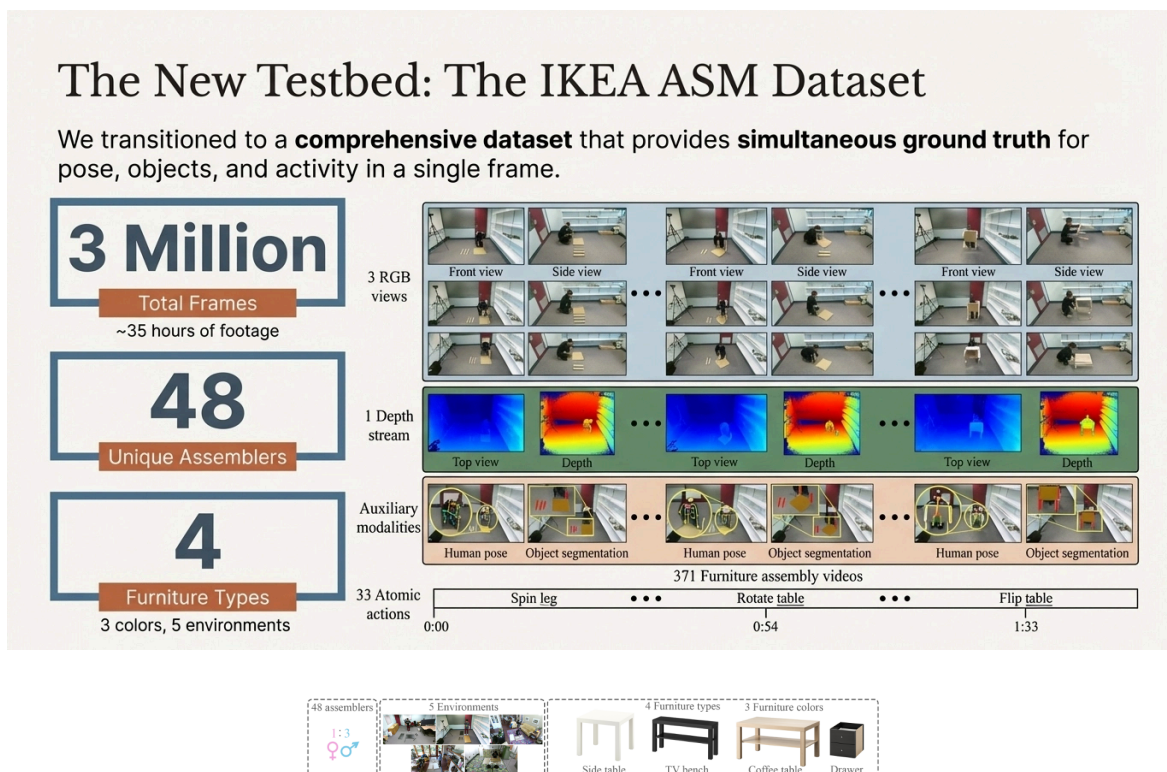


12th Meeting 3/15

it's been almost two months since our last weekly meeting. I'll be honest that I didn't meet our original goal of submitting a conference paper in January.

This multi-task learning, especially in a big data is simply demanded more effort in it than our initial schedule.



- **furniture part detection** problem with 7 object classes (table top, leg, shelf, side panel, etc).
- **2D human pose estimation** problem with 17 COCO keypoints (nose, eyes, shoulders, elbows, wrists, hips, knees, ankles).
- 33 atomic actions defined in the IKEA ASM dataset, such as "pick up leg", "attach shelf", "screw side panel", etc.

Why this matters.

- **Worker assistance and training**

The system can check if workers follow the correct steps, use the right parts, and maintain proper posture, enabling feedback, coaching, and automatic skill assessment.

- **Quality control and safety**

By understanding parts, pose, and actions together, the system can detect incorrect assembly steps early and monitor for unsafe postures or risky interactions.

- **Process analytics and optimization**

Automatic recognition of step-by-step actions and involved parts allows measurement of step durations, comparison between environments, and identification of bottlenecks.

- **Foundation for robotics / human-robot collaboration**

Detailed labels of how humans assemble objects (where they move, what they touch, what action they perform) provide valuable supervision for future robots that learn from human demonstrations.

Why a multi-task model (not just action recognition).

- **Detection only:** knows which parts are present, but not what the worker is doing with them.
- **Pose only:** knows body configuration, but not which object is manipulated or what stage of assembly it is.
- **Action only** (from raw image): may rely on misleading visual cues, lacks explicit understanding of parts or body pose, and is harder to interpret.

By sharing one backbone and learning three heads (det, pose, action), the model can:

- Use pose and parts as **strong, structured cues** for action recognition.
- Learn a **richer shared representation** than separate single-task models.
- Provide more interpretable outputs (we can inspect boxes and keypoints, not just a class label).

THE MOST IMPORTANT THING → It can be used in a CCTV to do a fully end-to-end analysis, from what they are doing, what the job is, how many parts they are working on, etc. For our system

The problem of this paper

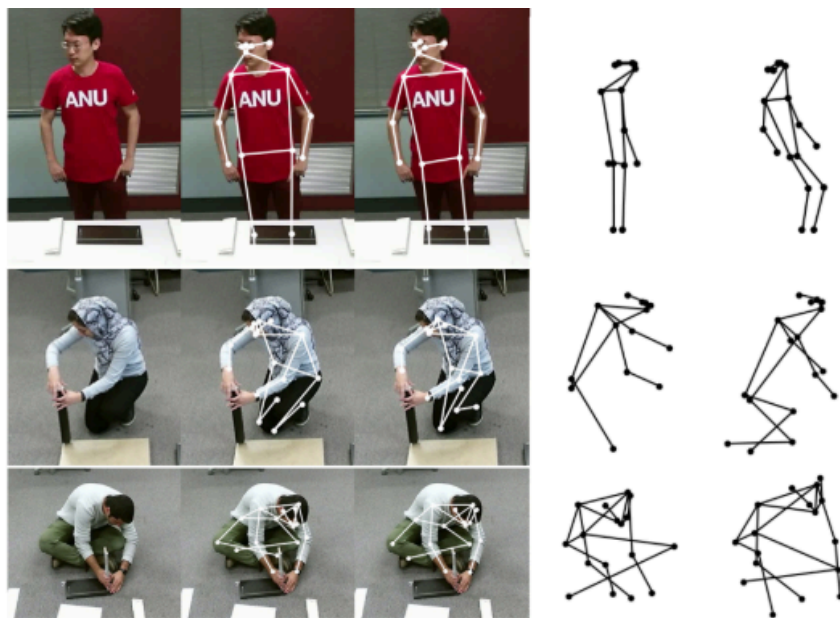


Figure 6: Qualitative human pose results. From left to right: sample image, 2D ground-truth, 2D Mask R-CNN prediction, 3D pseudo-ground-truth (novel view), and 3D VIBE prediction (novel view). The middle row shows an example where the 3D pseudo-ground-truth is incomplete, and the bottom row, shows a partial failure case for the predictions.

proaches, including self-occlusions, occlusions from furniture, baggy clothing, long hair, and human distractors in the background. We also obtain pseudo-ground-truth 3D annotations by fine-tuning a Mask R-CNN [22] 2D joint detector on the labeled data, and triangulating the detections of the model from the three calibrated camera views. As a verification step, the 3D points are backprojected to 2D and are discarded if more than 30 pixels from the most confident ground-truth annotations. The reprojection error of the true and pseudo ground-truth annotations is 7.12 pixels on the train set (83% of ground-truth joints detected) and 9.14 pixels on the test set (53% of ground-truth joints detected).

To evaluate the performance of benchmark 2D human pose approaches, we perform inference with existing state-

IKEA ASM dataset paper (Ben-Shabat et al.)

<https://arxiv.org/pdf/2007.00394>

Why we need it

- It is the **primary reference** for the dataset I use: it defines the atomic actions, the annotation protocols, and the official train/test splits.
- It provides **baseline results** for action recognition, pose, and object segmentation on IKEA assembly videos.
- This lets me:
 - Use their **atomic action taxonomy** and indexing files (`train_cross_env.txt`, `test_cross_env.txt`).
 - Compare my **multi-task model** against their **separate baselines** conceptually.

High-level idea

- They consider **actions, objects, and pose**, but mainly train **separate** models:
 - an action model on videos, a pose model, and an instance segmentation model.
 - They do not explore pose-conditioned modulation (FiLM) for action recognition.
 - Directly compare against their action baseline on the same splits once my training is stable.

What can we compare

1. Action

Method	Frame acc.			
	top 1	top 3	macro	mAP
ResNet18 [23]	27.06	55.14	21.95	11.69
ResNet50 [23]	30.38	56.1	20.03	9.47
C3D [69]	45.73	69.56	32.48	21.98
P3D [52]	60.4	81.07	45.21	29.86
I3D [8]	57.57	76.55	39.34	28.59

Table 2: Action recognition baseline frame-wise accuracy, macro-recall, and mean average precision results.

2. Pose2D

Method	Input	Train set			Test set		
		MPJPE↓	PCK↑	AUC↑	MPJPE↓	PCK↑	AUC↑
OpenPose-pt [7]	Image	17.3	46.9	78.1	16.5	46.7	77.8
OpenPose-ft [7]	Image	11.8	57.8	87.7	13.9	52.6	85.6
MaskRCNN-pt [22]	Image	15.5	51.9	78.2	16.1	51.5	79.2
MaskRCNN-ft [22]	Image	7.6	77.6	92.1	11.5	64.3	87.8
STAF-pt [53]	Video	21.4	41.8	75.3	19.7	41.1	75.4

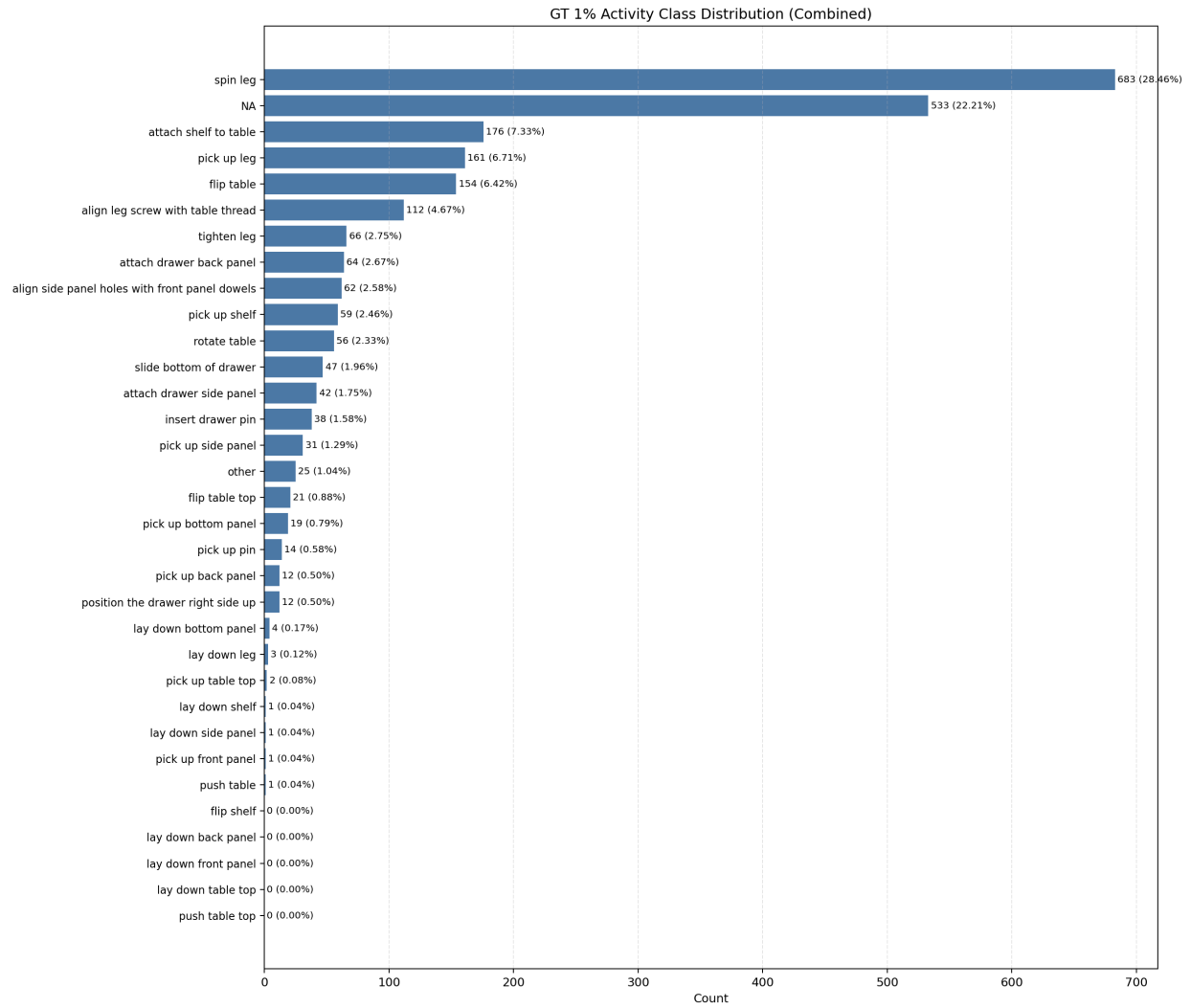
Table 7: 2D human pose results. The Mean Per Joint Position Error (MPJPE) in pixels and the Percentage of Correct Keypoints (PCK) @ 10 pixels (0.5% image width) are reported. Pretrained models are denoted ‘pt’ and models fine-tuned on the training data are denoted ‘ft’.

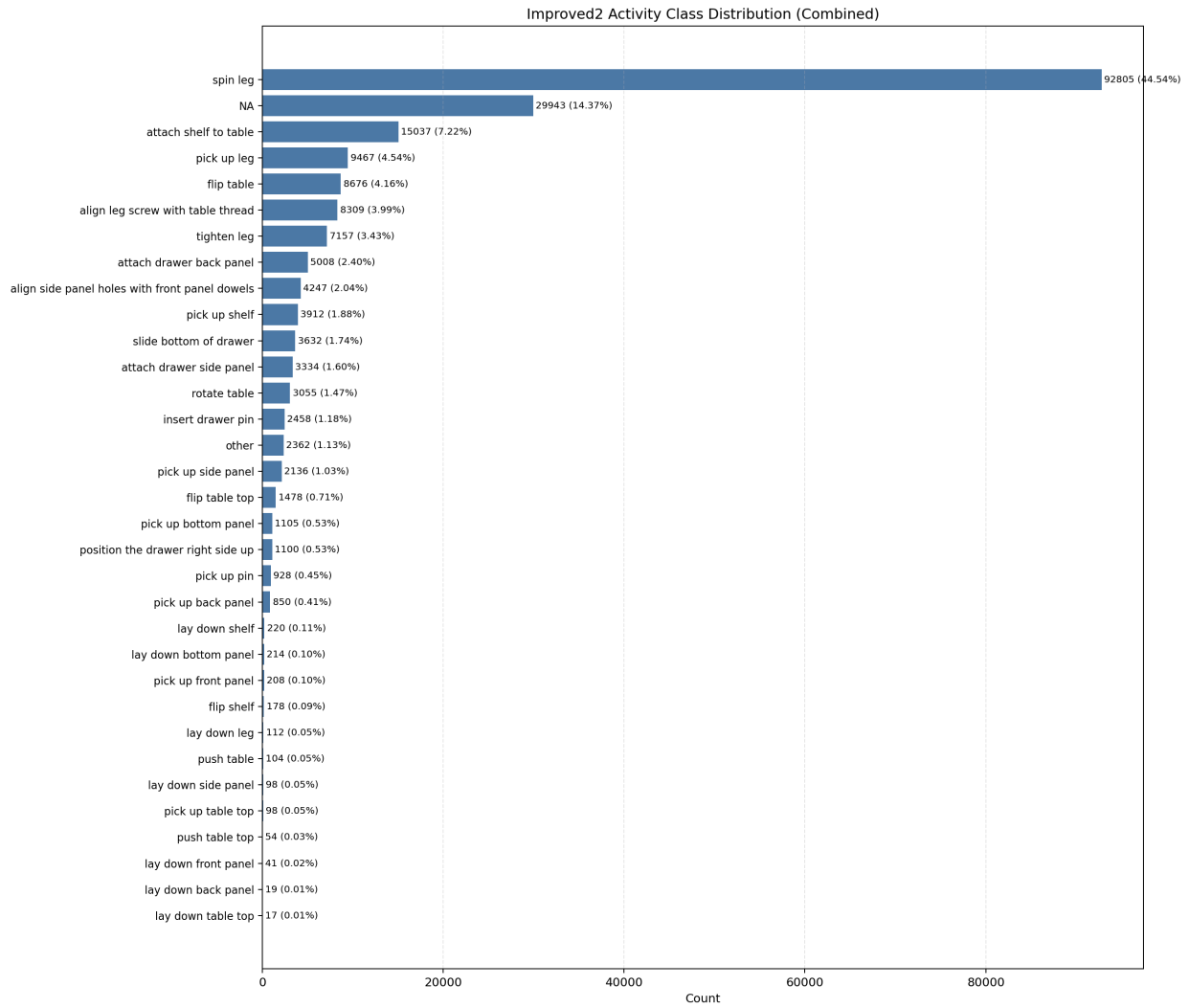
3. Detection

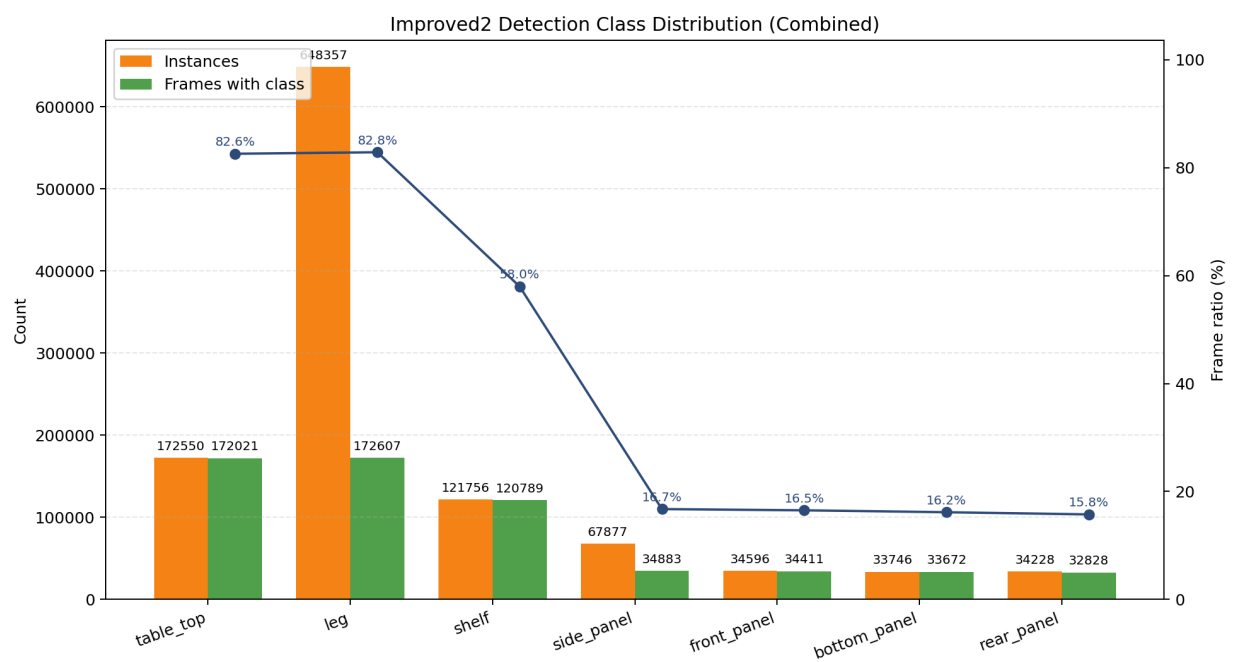
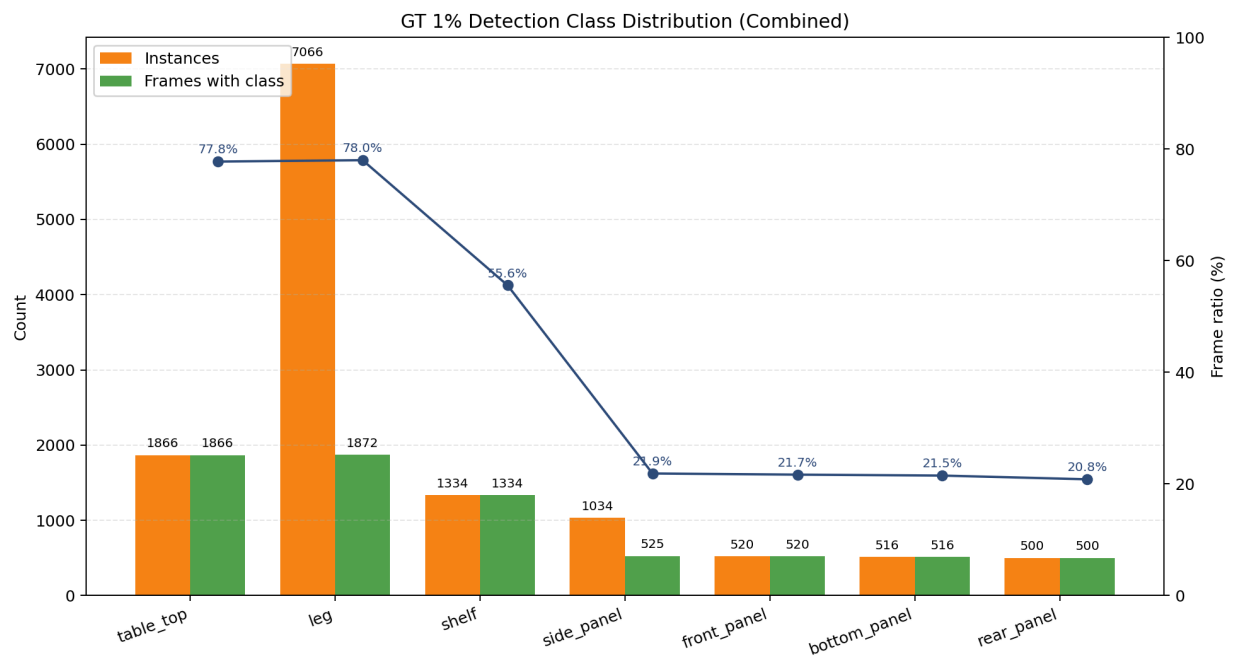
Feature Extractor	Annotation Type	AP	AP50	AP75	table-t	leg	shelf	side-p	front-p	bottom-p	rear-p
ResNet-50-FPN	mask	58.1	77.2	64.2	80.8	59.8	68.9	32.8	50.0	66.0	48.3
ResNet-101-FPN	mask	62.1	82.0	68.0	84.4	71.6	67.5	33.5	53.7	70.2	54.0
ResNeXt-101-FPN	mask	65.9	85.3	73.2	87.6	71.2	76.0	44.3	52.6	73.4	56.2
ResNet-50-FPN	bbox	59.5	77.7	68.9	77.3	63.5	64.7	41.0	60.1	61.8	48.5
ResNet-101-FPN	bbox	64.6	81.8	72.8	84.9	75.6	66.0	42.4	61.6	68.1	53.3
ResNeXt-101-FPN	bbox	69.5	86.4	78.9	89.4	76.8	73.7	53.3	65.8	68.7	59.0

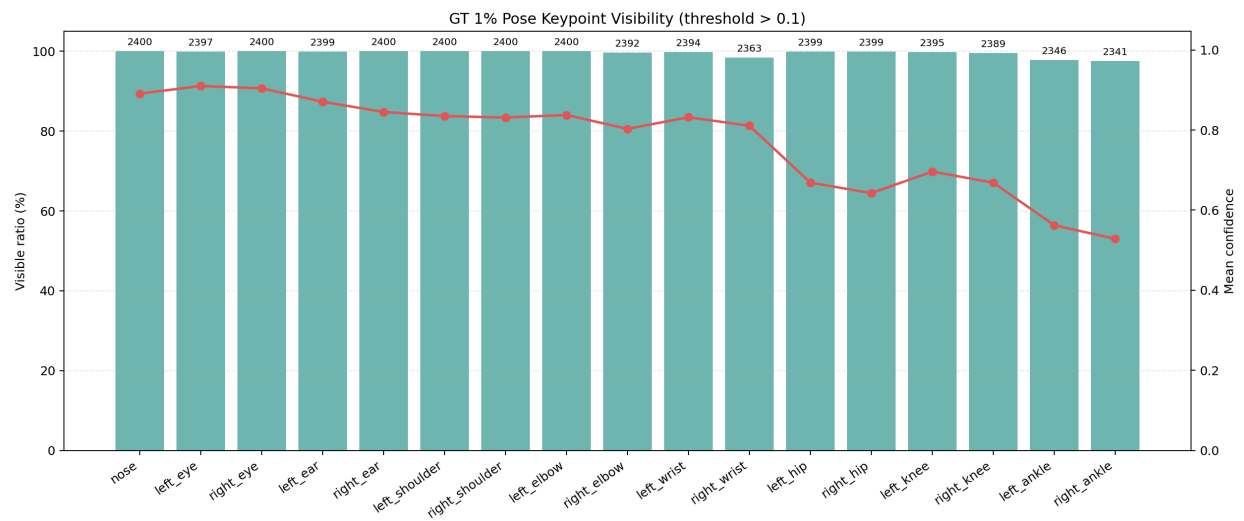
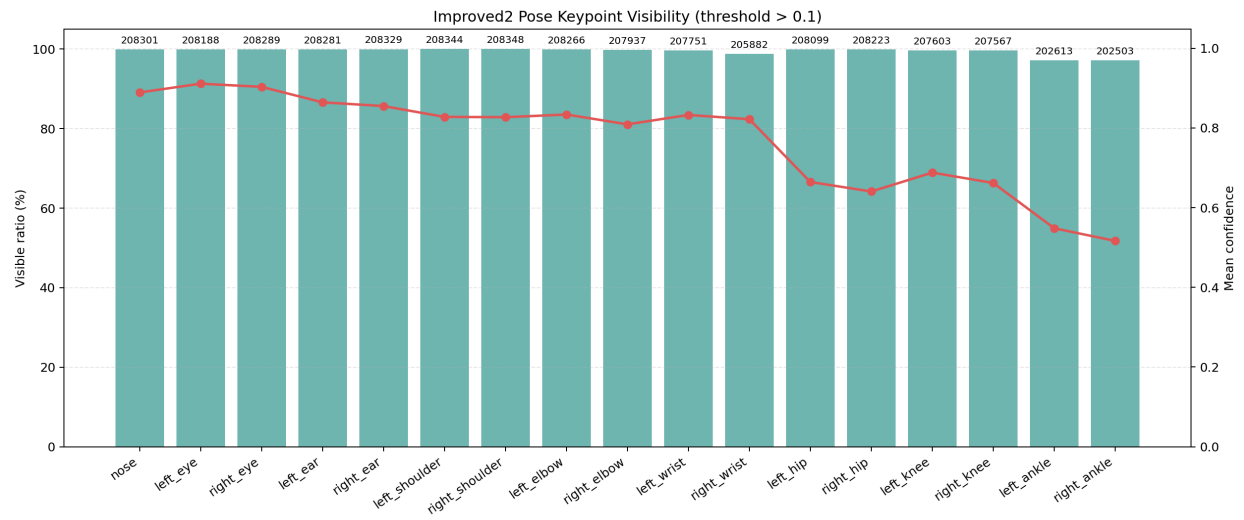
Table 4: Evaluating the effect of backbone architecture of Mask R-CNN in furniture part instance segmentation.

The problem of 1% ground truth is the distribution:









Since the manual annotation only covers 1% of the whole dataset, we propose to extract pseudo-ground-truth automatically. To this end, we train 12 different Mask R-CNNs with a ResNet50-FPN backbone to overfit on subsets of the training set that cover similar environments and furniture. We show that to achieve manual-like annotations with more accurate part boundaries, training the models with PointRend [30] as an additional head is essential. Figure 4 compares the automatically generated pseudo-ground-truth with and without the PointRend head. To evaluate the effectiveness of adding pseudo-ground-truth, we compare the Mask R-CNN trained with ResNet50-FPN with 1% annotated data (i.e., manual annotations) and 20% annotated data

(a) Influence of adding Pseudo GT						
Setting	AP_{segm}	$AP50_{segm}$	$AP75_{segm}$	AP_{box}	$AP50_{box}$	$AP75_{box}$
Manual GT	58.1	77.2	64.2	59.5	77.7	68.9
Manual + Pseudo GT	60.1	77.7	66.1	62.6	77.8	69.9

(b) Influence of PointRend head						
Setting	AP_{segm}	$AP50_{segm}$	$AP75_{segm}$	AP_{box}	$AP50_{box}$	$AP75_{box}$
Without PointRend	58.1	77.2	64.2	59.5	77.7	68.9
With PointRend	61.4	80.9	67.0	63.9	82.2	73.2

(c) Color-based Evaluation						
Colors	AP_{segm}	$AP50_{segm}$	$AP75_{segm}$	AP_{box}	$AP50_{box}$	$AP75_{box}$
White	55.5	76.2	60.6	57.3	76.5	65.5
Black	57.8	74.8	64.4	58.5	75.4	67.6
Oak	62.9	82.1	69.5	64.5	82.3	75.8

(d) Environment-based Evaluation						
Environments	AP_{segm}	$AP50_{segm}$	$AP75_{segm}$	AP_{box}	$AP50_{box}$	$AP75_{box}$
Env1 (Family Room)	47.1	63.0	53.5	49.9	65.6	58.4
Env2 (Office)	64.4	85.0	70.7	64.8	84.3	74.6



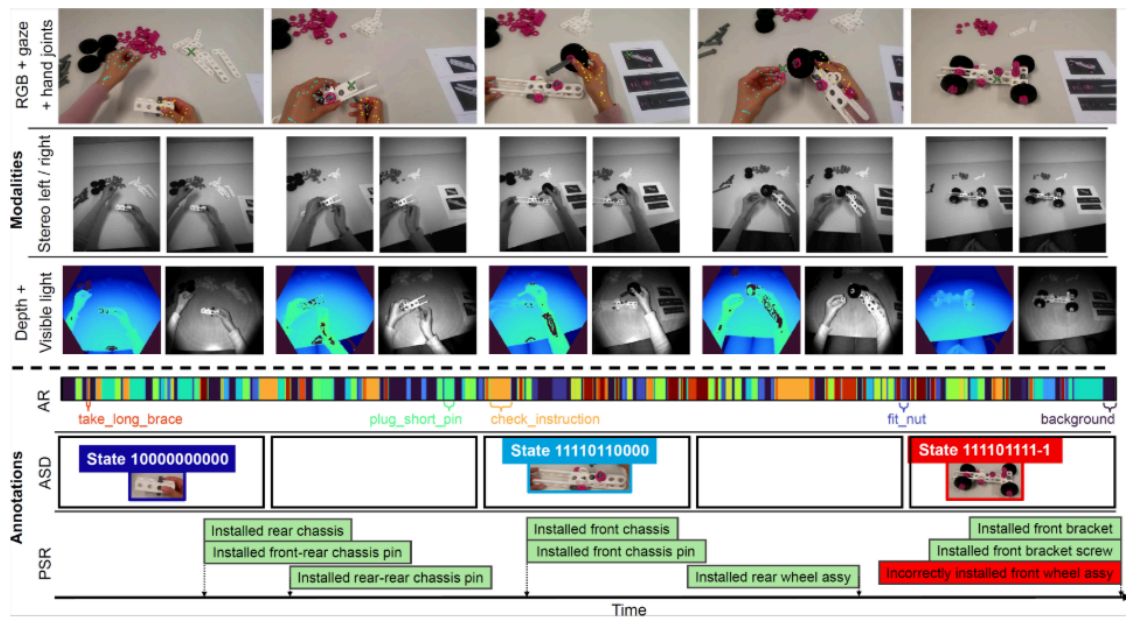
Figure 5: Illustration of part instance segmentation failure cases. (Top row) Mask R-CNN fails to correctly classify different panels of the drawer due to high similarity. (Bottom row) Mask R-CNN incorrectly segments part of the working surface as a furniture part (e.g., shelf) leading to considerable false positives.

Alternatives Dataset

1. Industreal

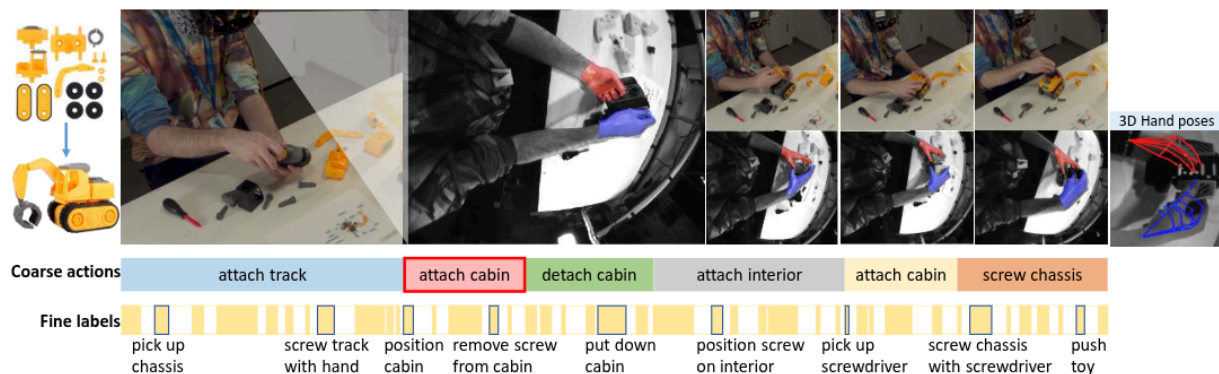
Annotations

IndustReal is specifically created to stimulate research on the novel procedure step recognition task. However, to broaden the possible uses of the dataset, we provide action recognition and object detection (assembly state detection) labels alongside the PSR annotations:



Samples from a clip in the IndustReal dataset, demonstrating the modalities and annotations for all three tasks. Gaze is indicated by the cross, detected hand joints by the dots. AR: action recognition, ASD: assembly state detection, PSR: procedure step recognition.

2. Assembly101



3. HA4M

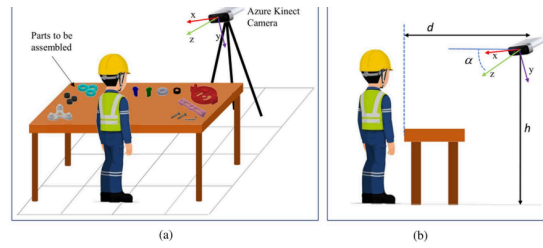
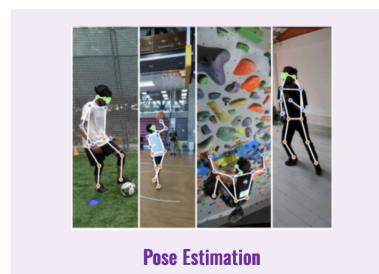
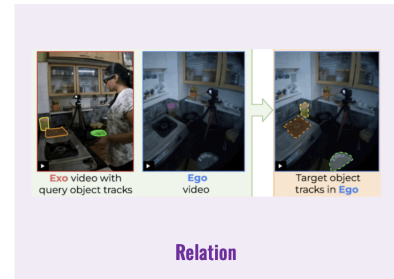
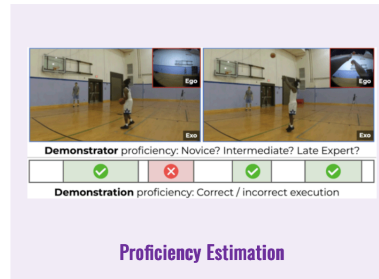
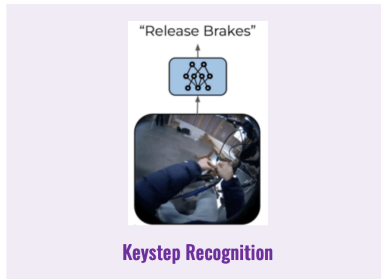


Fig. 2 Sketch of the acquisition setup: (a) a Microsoft® Azure Kinect is placed in front of the operator and the table where the components are spread over; (b) setup specifications.



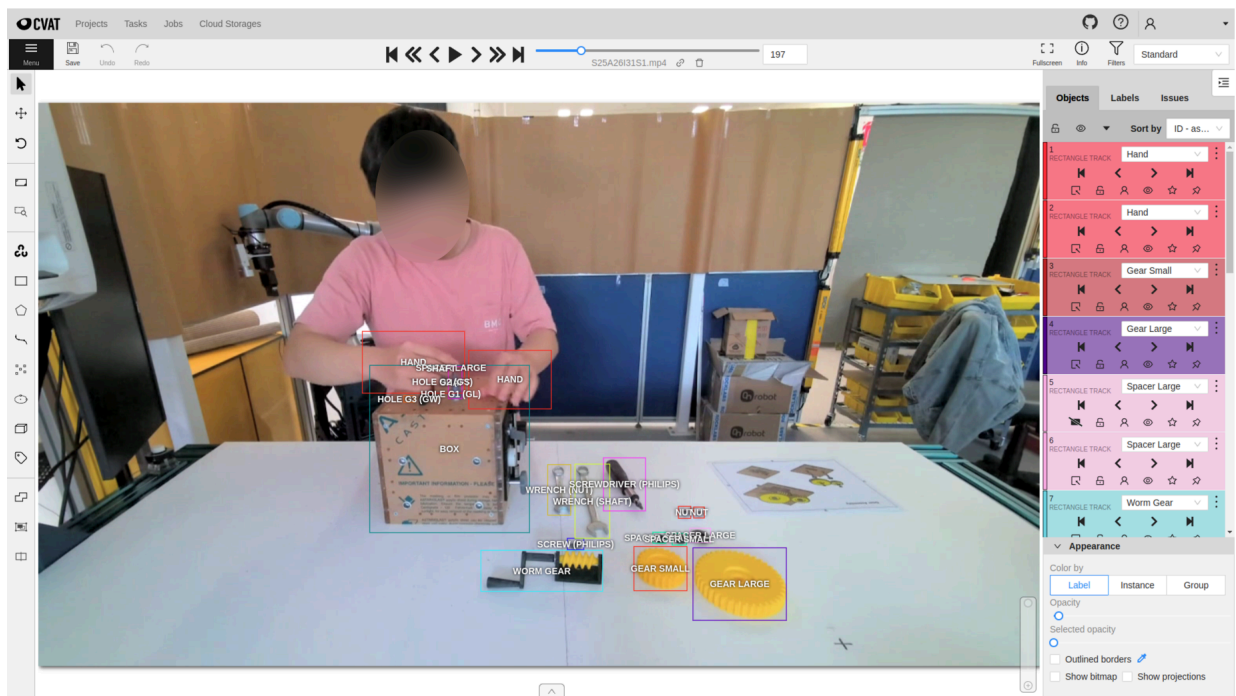
4. Ego Exo4D

BENCHMARKS

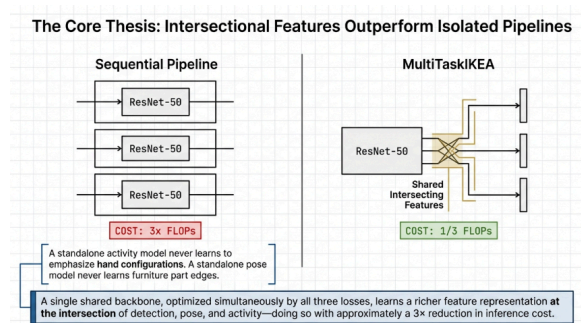


5. Ha-VID

For spatial annotations, we use CVAT, a video annotation tool, to label bounding boxes for subjects, objects and tools frame-by-frame.



baseline methods and model



- The one reason of why i use resnet50+FPN is because its performance, and already tested in this dataset → I will explain the whole end to end of the model if i already successfully gather the result from this dataset by next week
- As well as the rough chapter 1-3 for the papers by two weeks.

Afternoon next wednesday Third period 4th period thursday



- Performance better → Identify the problem → multitask

capture context → transformer

- Breakdown example / concrete example of it
- Structure of the model

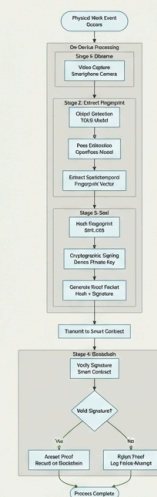
Roadmap Step 4: Blockchain Implementation

Long-Term Vision

WorkerNet is the critical AI foundation required to enable our ultimate goal: the Proof of Physical Work (POPW) protocol.

The POPW Pipeline:

1. WorkerNet generates a continuous stream of verified actions in real-time.
2. We extract a Spatiotemporal Fingerprint (Worker ID, pose trajectories, objects handled).
3. The unique hash is cryptographically signed and submitted to a smart contract.
4. Smart contract verifies the physical work and executes automated payment.



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